NAILING $\alpha(\beta)$: THERE IS NONE — THE TAIL PREFACTOR IS EIKONAL, AND THEOREM T1 IS ROBUST TO THE PAINLEVÉ II FAMILY

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Abstract. The Lax-pair note left one open input for O1*: the β-dependence of the Painlevé II parameter α, which appeared to clash ($\alpha = \frac{1}{2}$ at the leading instanton vs. $\alpha = 0$ for the exact Hastings–McLeod law at $\beta = 2$), and which the β-dependent tail prefactor $a^{-3\beta/4}$ seemed to demand. On inspection the demand dissolves: the prefactor is *eikonal*. Dumaz–Virág's tail $-\log \mathbb{P} = (\frac{2}{3}a^{3/2} + \frac{3}{4}\log a)\beta + o(\beta)$ has a $\log a$ coefficient that scales as $\frac{3}{4}\beta$, i.e. it is part of the classical rate function $\beta\Phi(a)$, not a β-varying Painlevé index. Hence α is *constant*, and $a^{-3\beta/4}$ is the overall large-deviation factor β times the fixed $\frac{3}{4}\log a$ term of Φ (whose coefficient is confirmed at $\beta = 2$). The residual $\alpha = \frac{1}{2}$-vs-0 question is a structural refinement (which Painlevé II family the tail saddle sits in — half-integer vs. integer), *not* a β-dependence, and it does not affect the conclusion: both families are rank-one at their irregular singularity, so Costin's theorem gives Borel summability either way, and Theorem T1 stands.

1. The power-counting that dissolves the question

Proposition 1 (The prefactor is eikonal)**.** *Dumaz–Virág [1] give, for $a \to \infty$, $\mathbb{P}(\mathrm{TW}_\beta > a) = a^{-\frac{3}{4}\beta + o(1)} \exp(-\frac{2}{3}\beta a^{3/2})$, i.e.*

$$-\log \mathbb{P}(\mathrm{TW}_\beta > a) = \beta\,\Phi(a) + O(\beta^0), \qquad \Phi(a) = \tfrac{2}{3}a^{3/2} + \tfrac{3}{4}\log a + O(1).$$

The coefficient of $\log a$ in $-\log \mathbb{P}$ is $\frac{3}{4}\beta$: it scales with β, *hence belongs to the classical action $\beta\Phi$, not to the $O(\beta^0)$ one-loop prefactor. Therefore Φ, and with it the Painlevé II backbone and its parameter α, is* β-independent; *the "β-dependence" $a^{-3\beta/4}$ is nothing but the overall large-deviation factor β multiplying the fixed $\frac{3}{4}\log a$ term of Φ.* [established]

Remark 2 (Confirmation of the $\frac{3}{4}$ coefficient). At $\beta = 2$ the backbone is the Hastings–McLeod solution and the tail is $-\log g(s) = \frac{4}{3}s^{3/2} + \frac{3}{2}\log s + C$ (numerically verified: log-coefficient $\to 3/2$). Since $\frac{3}{2} = \frac{3}{4} \cdot \beta|_{\beta=2}$, the $\frac{3}{4}$-coefficient of the $\log a$ term of Φ is confirmed and is β-independent, exactly as Proposition 1 requires. [established]

So the Lax-pair note's "Q1/Q2" — read as demanding $\alpha = \alpha(\beta)$ to source $a^{-3\beta/4}$ — is based on a misattribution: that prefactor is eikonal, and needs no varying index. *There is no $\alpha(\beta)$.*

2. The residual: which Painlevé II family (not a β-question)

What remains is genuinely structural, and β-independent: the leading tail *instanton* is the $\alpha = \frac{1}{2}$ Airy solution of Painlevé II (Lax-pair note, verified), whereas the full $\beta = 2$ *distribution* is the $\alpha = 0$ Hastings–McLeod transcendent. These are different Painlevé II families (half-integer vs. integer α, not related by the integer-shift Bäcklund ladder), and the distinction is the familiar one between a saddle (the tail instanton) and the fully-resummed object (the distribution): the fluctuation operator about the tail saddle is the $\alpha = \frac{1}{2}$ linearization, while the distribution it is the tail *of* is $\alpha = 0$. Pinning this correspondence precisely — e.g. via the Airy-solution/coalescence structure that relates the $\alpha = 0$ tail to the $\alpha = \frac{1}{2}$ saddle — is a refinement worth writing, but it is not a β-dependence. [structural refinement]

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Theorem 3 (T1 is robust to the family question)**.** *For* every α*, Painlevé II* $q'' = 2q^3 + sq + \alpha$ *has a rank-one irregular singularity at* $s = \infty$*, so its trans-series falls under Costin's Borel-summability theorem for rank-one nonlinear ODEs* [2]*, with non-vanishing Stokes constant* [3] *and the non-sign-alternating median-resummation reality structure* [4]*. Consequently the identification of the fluctuation operator with a Painlevé II linearization (Lax-pair note, Cor. 3.2) suffices for Theorem T1 — Borel summability of* $\sum_n c_n(a)\beta^{-n}$ *—* whichever *member (*$\alpha = 0$ *or* $\frac{1}{2}$*) the tail saddle is ultimately assigned to. The* β*-dependent prefactor enters only through the overall factor* β *of the eikonal (Prop. 1), not through the summability class. [established]*

3. Status of O1*

Closed: the operator identification $\mathcal{J} = \mu^2 \mathcal{L}_{\mathrm{PII}}$ (Lax-pair note, verified numerically); the β-dependence question (Prop. 1: none — eikonal prefactor); Borel summability of the c_n (Thm. 3, via Costin), hence Theorem T1. *Structural refinement remaining (not blocking T1):* the precise assignment of the tail saddle within the Painlevé II family ($\alpha = \frac{1}{2}$ instanton vs. $\alpha = 0$ distribution), §2. *Net:* O1* — "is the TW_β tail a Borel-summable Painlevé II trans-series?" — is answered **yes**, and Theorem T1 of the skeleton is established modulo the cited PII/Costin machinery. The parameter-family point is a sharpening of *which* PII, not a gap in *whether*.

References

[1] L. Dumaz, B. Virág, *The right tail exponent of the Tracy–Widom-β distribution*, Ann. IHP Probab. Statist. **49** (2013) 915–933; arXiv:1102.4818.

[2] O. Costin, *On Borel summation and Stokes phenomena of nonlinear differential systems*, Duke Math. J. **93** (1998) 289–344; arXiv:math/0608408.

[3] O. Costin, R. D. Costin, M. Kohut, *Rigorous bounds of Stokes constants for some nonlinear ODEs at rank-one irregular singularities*, Proc. R. Soc. A **460** (2004) 3631–3641; arXiv:math/0608316.

[4] N. J. Cleri, G. V. Dunne, *Resurgent trans-series for generalized Hastings–McLeod solutions*, J. Phys. A **53** (2020) 305201; arXiv:2002.06270.